

demand estimation and market analysis: air fryers

Notebook 2 - demand estimation Role: data scientist - turning cleaned data into a model that can be used for prediction and interpretation.

The demand model is:

$$\log(s_{bt}) = \alpha_0 + \alpha_t + \gamma_b + \beta_{\text{price}} p_{bt} + \beta_{\text{rating}} r_{bt} + \sum_{\ell} \beta_{\ell} x_{b\ell t} + \epsilon_{bt}$$

where year dummies α_t absorb the year specific denominator of the multinomial logit share equation.

```
import pandas as pd
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import statsmodels.api as sm
import warnings
warnings.filterwarnings('ignore')

df = pd.read_csv('air_fryers_clean_brand_year.csv')
char_cols = ['compact_share', 'dual_basket_share', 'oven_style_share', 'rotisserie_share', 'window_share']

brand_dummies = pd.get_dummies(df['brand'], drop_first=True, prefix='brand').astype(float)
year_dummies = pd.get_dummies(df['year'], drop_first=True, prefix='year').astype(float)

X = pd.concat([df[['avg_price', 'avg_rating']+char_cols].astype(float),
               brand_dummies, year_dummies], axis=1)
X = sm.add_constant(X)
y = df['log_brand_share'].astype(float)

model = sm.OLS(y, X).fit()
print(model.summary())
```

```

OLS Regression Results
=====
Dep. Variable:      log_brand_share    R-squared:                0.763
Model:              OLS               Adj. R-squared:           0.600
Method:             Least Squares      F-statistic:              4.680
Date:               Mon, 04 May 2026    Prob (F-statistic):       9.24e-05
Time:               13:52:16           Log-Likelihood:           -36.451
No. Observations:   50                AIC:                     114.9
Df Residuals:       29                BIC:                     155.1
Df Model:           20
Covariance Type:    nonrobust
=====
                    coef    std err          t      P>|t|      [0.025    0.975]
-----
const              -13.3049      16.283     -0.817     0.421    -46.608     19.998
avg_price           -0.0377       0.011     -3.378     0.002    -0.060    -0.015
avg_rating          0.2875       4.152     0.069     0.945    -8.204     8.779
compact_share       9.8153       3.637     2.699     0.011     2.378    17.253
dual_basket_share  -9.5097      34.030    -0.279     0.782   -79.109    60.089
oven_style_share    1.9418       1.506     1.290     0.207    -1.138     5.021
rotisserie_share    -5.6741       4.343    -1.307     0.202   -14.556     3.207
window_share       12.8803       6.467     1.992     0.056    -0.347    26.107
brand_cosori        2.5519       2.291     1.114     0.274    -2.133     7.237
brand_cuisinart     6.4224       2.465     2.606     0.014     1.381    11.464
brand_dash          0.1767       1.422     0.124     0.902    -2.732     3.086
brand_gowise usa    3.9390       2.105     1.872     0.071    -0.365     8.243
brand_instant_pot   4.6263       1.932     2.395     0.023     0.676     8.577
brand_ninja         5.8387       2.590     2.254     0.032     0.541    11.136
brand_nuwave        3.5449       1.910     1.856     0.074    -0.362     7.451
brand_oster         3.9281       2.129     1.845     0.075    -0.426     8.282
brand_ultrean       0.9424       1.756     0.537     0.596    -2.650     4.535
year_2020           0.1191       0.334     0.357     0.724    -0.564     0.802
year_2021           0.0419       0.348     0.120     0.905    -0.670     0.754
year_2022          -0.0989       0.351    -0.282     0.780    -0.817     0.619
year_2023          -0.0033       0.353    -0.009     0.993    -0.725     0.718
=====
Omnibus:            0.392    Durbin-Watson:           2.341
Prob(Omnibus):      0.822    Jarque-Bera (JB):         0.176
Skew:               -0.145    Prob(JB):                 0.916
Kurtosis:           2.998    Cond. No.                 4.94e+04
=====

```

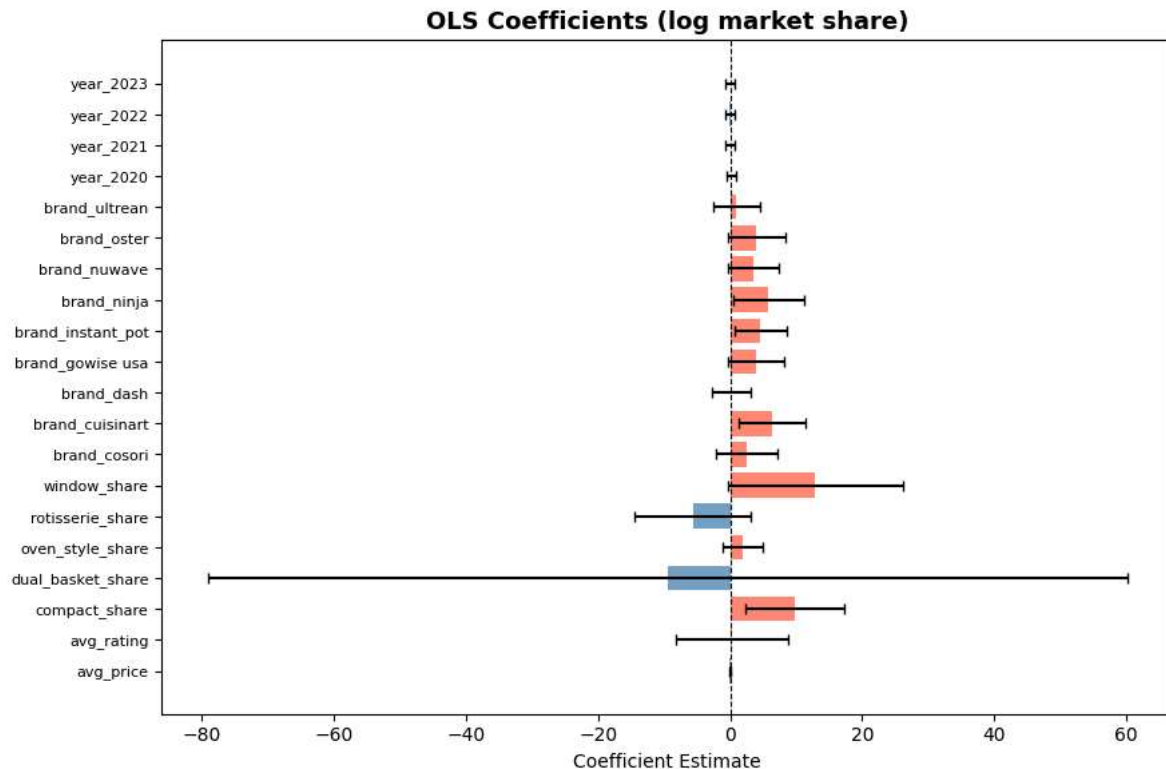
Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 4.94e+04. This might indicate that there are strong multicollinearity or other numerical problems.

2.1. Coefficient plot

```
coef = model.params.drop('const')
conf = model.conf_int().drop('const')
fig, ax = plt.subplots(figsize=(9,6))
y_pos = range(len(coef))
ax.barh(list(y_pos), coef.values,
        xerr=[coef.values - conf[0].values, conf[1].values - coef.values],
        color=['steelblue' if v < 0 else 'tomato' for v in coef.values],
        alpha=0.75, capsize=3)
ax.axvline(0, color='black', linewidth=0.8, linestyle='--')
ax.set_yticks(list(y_pos)); ax.set_yticklabels(coef.index, fontsize=8)
ax.set_title('OLS Coefficients (log market share)', fontsize=13, fontweight='bold')
ax.set_xlabel('Coefficient Estimate')
plt.tight_layout(); plt.show()
```



questions

1. what is the estimated price coefficient?

The estimated price coefficient $\hat{\beta}_{\text{price}}$ is **-0.0377** ($p = 0.0021$).

2. is it negative? why is that important?

Yes it is negative and that is critically important because it confirms the model is economically sensible: higher prices reduce demand, which is the fundamental prediction of standard consumer choice theory. If the coefficient were positive the entire cost and markup calculation would be invalid because it would imply that price increases lead to higher market shares which contradicts the basic logic of competitive markets. the negative sign is also required mathematically for the derived demand slope $\hat{s}'(p)$ to be negative, which in turn makes the inferred unit cost formula produce economically meaningful positive markups.

3. which product features are associated higher demand? compact designs has the largest and most statistically significant positive effect (coefficient ≈ 9.8 , $p = 0.011$), meaning brands with a higher share of compact products tend to have higher market shares. Window products also show a positive and marginally significant effect (coefficient ≈ 12.9 , $p = 0.056$). Rotisserie products show a negative coefficient (≈ -5.7) though it is not statistically significant at conventional levels. Dual basket and oven style have mixed small effects that are not statistically distinguishable from zero.

4. Which brand dummy coefficients are largest?

the reference brand is chefman (the first brand alphabetically, dropped by `drop_first = True`). relative to chefman, cuisinart has the largest brand dummy coefficient at about 6.42, followed by ninja at 5.84 and instant pot at 4.63. This means that controlling for price, ratings, and product characteristics, these brands have substantially high demand than chefman. It is important to note these coefficients capture brand reputation, loyalty, and unobserved quality differences rather than anything we can directly observe.

5. Which year dummy coefficients are largest

The reference year is 2019. the year dummies are all small in magnitude and statistically insignificant (all p-values above 0.7) with 2020 being the largest at about 0.12. this means that after controlling for brand characteristics and prices, there was no statistically meaningful aggregate shift in market level demand across years within this cleaned top 10 sample.

6. what is the model's R^2 ?

the model's R^2 is 0.763, meaning the model explains about 76.3% of the variation in log market share. this is a fairly high fit for a cross sectional panel with only 50 observations, though the adjusted R^2 of 0.600 reflects the large number of dummy variables relative to observations. the high fit is partly driven by the brand fixed effects absorbing persistent cross brand differences in popularity.

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.